



Column Curves of Steel–Concrete Composite Members

Ben Kato

Faculty of Engineering, Toyo University, 2100 Kujirai, Kawagoe City 350, Japan

(Received 4 September 1995; revised version received 25 April 1996; accepted 17 May 1996)

ABSTRACT

Buckling strength curves of centrally loaded composite columns were investigated based on the available test results. Concrete-filled circular and square steel tubes and concrete-encased steel H-section members were covered. It is demonstrated that the design formula given by ISO standard for steel structures can be applied to these composite columns with appropriate modifications for each type of column. Copyright © 1996 Elsevier Science Ltd

NOTATION

A	Sectional area in general
A_c	Area of concrete
A_s	Sectional area of steel section
a_r	Area of longitudinal reinforcing bars
B	Width of square hollow section, or coefficient
E	Modulus of elasticity of steel
E_c	Modulus of elasticity of concrete
f_c	Compressive strength of concrete
g	Reduction coefficient
g_c	Reduction coefficient obtained by test
I_c	Moment of inertia of concrete
I_r	Moment of inertia of longitudinal reinforcing bars
I_s	Moment of inertia of steel section
kL	Effective column length
N_c	Buckling strength of centrally loaded column
N_{cal}	Calculated buckling strength

N_{cr}	Eulerian elastic buckling strength
N_{test}	Experimental buckling strength
N_y	Squash load
t	Wall thickness of hollow section
ϕ	Diameter of circular hollow section
$\bar{\lambda}$	Relative slenderness
σ_y	Yield stress of steel
σ_{yr}	Yield stress of longitudinal reinforcing bars

1 BUCKLING STRENGTH OF CENTRALLY LOADED COMPOSITE COLUMNS

The buckling strength of centrally loaded columns, N_c , could be given by the general form as

$$N_c = gN_y \quad (1)$$

in which N_y is the squash load (maximum compressive strength which is not influenced by stability), and g is the reduction coefficient given in terms of the relative slenderness $\bar{\lambda}$. The relative slenderness $\bar{\lambda}$ is defined as

$$\bar{\lambda} = \sqrt{\frac{N_y}{N_{cr}}} \quad (2)$$

in which, $N_{cr} = \pi^2 D / L^2$ is the Eulerian elastic buckling strength, and $D = (EI)$ is the elastic flexural rigidity of a column. $\bar{\lambda}$ is related to the traditional slenderness, $\lambda = L/i$, as follows: $N_{cr} = A\pi^2 E / \lambda^2$ (A is the sectional area), and if the slenderness at which the Eulerian buckling stress attains to yield stress is denoted as λ_y , namely $N_y = A\sigma_y = A\pi^2 E / \lambda_y^2$, then eqn (2) is written as

$$\bar{\lambda} = \frac{\lambda}{\lambda_y} \quad (3)$$

Thus, the relative slenderness, $\bar{\lambda}$, is the slenderness normalized by λ_y (see Fig. 1). The actual column strength is lower than the ideal Eulerian buckling strength due to material and geometrical imperfections as shown in Fig. 1 by bold line and the reduction coefficient, g , is the actual column strength reduction mode from the squash load, N_y , in relation to $\bar{\lambda}$.

In ISO standard for steel structure,¹ N_y and g are given as follows:

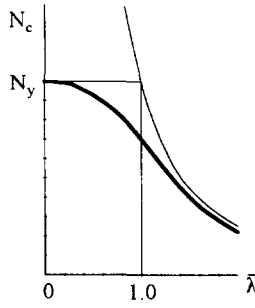


Fig. 1. Relative slenderness.

$$N_y = A_s \sigma_y \quad (4)$$

$$g = B(1 - \sqrt{1 - C}) \quad (5)$$

$$B = \frac{1 + \alpha(\bar{\lambda} - \bar{\lambda}_0) + \bar{\lambda}^2}{2\bar{\lambda}^2}, \quad C = (B\bar{\lambda})^{-2}$$

$$\bar{\lambda}_0 = 0.2$$

$$\alpha = \begin{cases} 0.21 & \text{for curve } a \\ 0.34 & \text{for curve } b \\ 0.49 & \text{for curve } c \end{cases}$$

$$\bar{\lambda} = \sqrt{\frac{N_y}{N_{cr}}} = \frac{kL}{\pi} \sqrt{\frac{A\sigma_y}{EI_s}}$$

in which

A_s : sectional area of steel section;
 kL : effective column length;
 E : modulus of elasticity of steel;
 I_s : moment of inertia of steel section;
 σ_y : yield stress of steel.

Three column strength reduction modes a , b and c are provided depending on the degree of imperfection using the variable coefficient α . If one adopts this concept for the design of composite columns also, the squash load, N_y , and the relative slenderness, $\bar{\lambda}$, could be defined as

$$N_y = A_s \sigma_y + (a_t \sigma_{yr}) + A_c f_c \quad (6)$$

in which

A_c : area of concrete;
 f_c : compressive strength of concrete;
 a_r : area of longitudinal reinforcing bars for concrete-encased steel column if any;
 σ_{yr} : yield stress of longitudinal reinforcing bars for concrete-encased steel column if any.

$$\bar{\lambda} = \frac{kL}{\pi \sqrt{\frac{A_s \sigma_y + (a_r \sigma_{yr}) + A_c f_c}{E(I_s + I_r) + E_c I_c}}} \quad (7)$$

in which

I_r : moment of inertia of longitudinal reinforcing bars for concrete-encased steel column;
 I_c : moment of inertia of concrete;
 E_c : modulus of elasticity of concrete.

In the following, the strength prediction using eqn (6) and eqn (7) and the reduction coefficient, g , given in eqn (5) are compared with the available experimental results for concrete-filled steel tubular columns (circular and square) and concrete-encased steel columns (with and without longitudinal reinforcements), and the suitability of the prediction formulae are assessed.

2 CONCRETE-FILLED STEEL TUBULAR COLUMNS

In concrete-filled steel tubular columns, longitudinal reinforcing bars are generally not installed, therefore the squash load, N_y , and the relative slenderness, $\bar{\lambda}$, are written as

$$N_y = A_s \sigma_y + A_c f_c \quad (6')$$

$$\bar{\lambda} = \frac{kL}{\pi \sqrt{\frac{A_s \sigma_y + A_c f_c}{EI_s + E_c I_c}}} \quad (7')$$

2.1 Concrete-filled circular steel tube columns

Thirty-seven test results from Japan²⁻⁴ and 63 test results from abroad⁵ were collected. The variation of size and material properties covered in these tested specimens are 25.4–267.4 mm in tube diameter, 275–682 N/mm² in yield

stress of steel and 20.3–66.2 N/mm² in compressive strength of concrete. For test specimens on which the modulus of elasticity of concrete, E_c , were not measured, E_c was estimated by eqn (8):

$$E_c = 2.058 \times 10^4 \times \sqrt{\frac{f_c}{19.6}}, \text{ N/mm}^2. \quad (8)$$

The column strength evaluation consists of the selection of the squash load, N_y , and of the reduction coefficient, g . First of all, the configuration of the reduction mode, g , is surveyed. Namely, the experimental reduction coefficient, g_e , is obtained from eqn (1) as

$$g_e = \frac{N_{\text{test}}}{N_y} \quad (9)$$

in which, N_{test} is the maximum column strength obtained by test, and N_y is the squash load calculated by eqn (6'). They are compared with the formulated reduction coefficient, g , given by eqn (5) in Fig. 2. From this figure, it can be observed that the actual squash load is higher than the squash load given by eqn (6'). This may be caused by the increase of the compressive strength

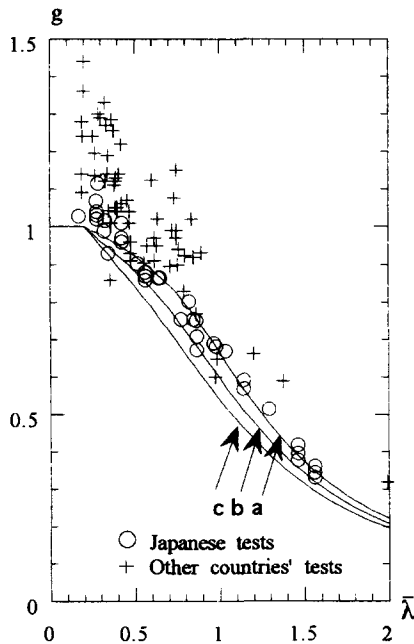


Fig. 2. Concrete-filled circular steel tube columns, $1.0 f_c$.

of in-filled concrete due to the effect of confinement, and through a parametric study it was concluded that concrete strength could be factored by 1.1, and the curve *b* could be used for the estimation of the reduction coefficient, *g*.

The results of this modification are summarized in Fig. 3 and Table 1. The correlation between the formulated reduction mode (curve *b*) and the experimental reduction coefficients is depicted in Fig. 3, and the comparison of test results, N_{test} , and the calculated column strength, N_{cal} , using $1.1 f_c$ and curve *b* is given in Table 1. The effect of confinement, namely the increase in concrete strength, cannot be represented by the simple term of $1.1 f_c$, but it depends on the strength ratio and the stiffness ratio of steel and concrete, and therefore some scattering of experimental reduction coefficient in smaller slenderness regions is inevitable.

2.2 Concrete-filled square steel tube column

Thirteen test results from Japan^{6,7} were collected but no test results from abroad were available. The yield stresses of steel are 412 and 431 N/mm², and the compressive strength of concrete varies from 24.0 to 32.8 N/mm². They are of moderate value. For test specimens on which the modulus of elasticity of concrete, E_c , were not measured, E_c was estimated by eqn (8). The experimental reduction coefficient, g_e , is estimated by eqn (9) in which

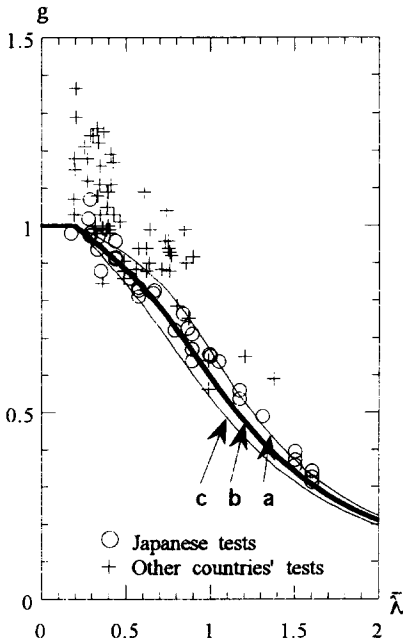


Fig. 3. Concrete-filled circular steel tube columns, $1.0 f_c$.

TABLE 1
Concrete-filled Steel Tubular Columns. Circular Hollow Sections ($1.1 f_c$)

No.	ϕ (mm)	t (mm)	kL (cm)	σ_y (N/mm ²)	f_c (N/mm ²)	$\bar{\lambda}$	N_{cal} (kN)	N_{test} (kN)	$\frac{N_{test}}{N_{cal}}$	Ref.
1	165.2	4.1	66.1	353	40.9	0.175	1599	1562	0.98	2
2			132.2			0.350	1516	1412	0.93	
3			198.2			0.525	1397	1372	0.98	
4			297.4			0.788	1171	1147	0.98	
5			396.5			1.051	902	1019	1.13	
6			495.6			1.313	675	782	1.16	
7	190.7	5.5	115.0	505	48.3	0.289	2821	3148	1.12	3
8			345.0			0.866	2101	2133	1.01	
9	267.4	6.4	160.0	461		0.280	4966	5188	1.04	
10			480.0			0.840	3568	3903	1.09	
11	190.7	5.5	115.0	505	55.9	0.335	2987	3063	1.02	
12			230.0			0.669	2532	2612	1.03	
13			345.0			1.000	1889	2059	1.09	
14	267.4	6.4	160.0	461		0.331	5280	5181	0.98	
15			320.0			0.662	4451	4534	1.02	
16			480.0			0.994	3323	3626	1.08	
17	114.3	3.1	67.0	348	53.5	0.288	896	898	1.00	4
18								904	1.01	
19								910	1.02	
20			102.0		53.4	0.439	842	888	1.05	
21								849	1.01	
22								845	1.00	
23			133.5		56.0	0.576	809	784	0.97	
24								773	0.96	
25								794	0.98	
26			204.0		57.5	0.891	643	617	0.96	
27								649	1.01	
28								688	1.07	
29			270.0	340	57.5	1.176	471	516	1.09	
30								537	1.14	
31								536	1.14	
32			337.0		63.1	1.506	345	401	1.16	
33								362	1.05	
34								379	1.10	
35			372.0		55.1	1.605	286	320	1.12	
36								293	1.02	
37								305	1.07	

TABLE 1 Continued

No.	ϕ (mm)	t (mm)	kL (cm)	σ_y (N/mm ²)	f_c (N/mm ²)	$\bar{\lambda}$	N_{cal} (kN)	N_{test} (kN)	$\frac{N_{test}}{N_{cal}}$	Ref.
Foreign tests										
1	139.7		91.4	682	38.4	0.361	2878	3047	1.06	5
2	139.7		304.8	682		1.204	1459	2002	1.37	
3	177.8		304.8	593		0.899	3113	4310	1.38	
4	139.7		91.4	682		0.361	2878	2598	0.90	
5	177.8		91.4	593		0.270	4587	5253	1.14	
6	177.8		91.4	593		0.270	4587	5525	1.20	
7	95.0		86.1	275	20.3	0.349	929	1019	1.10	
8			142.0			0.575	837	930	1.11	
9			198.1			0.803	710	903	1.27	
10			86.1	350	25.0	0.368	501	667	1.33	
11			142.0			0.607	445	583	1.31	
12			198.1			0.846	371	529	1.43	
13	215.9		222.0	292	22.8	0.395	1524	1650	1.08	
14				292	29.8	0.413	1750	2264	1.29	
15				392	22.8	0.428	2055	2442	1.19	
16				350	29.8	0.426	2252	2869	1.27	
17	95.0		203.2	338	24.0	0.855	355	463	1.30	
18	120.9		104.9	312	21.1	0.332	537	721	1.34	
19					24.2	0.337	663	854	1.29	
20			231.1		21.1	0.731	509	636	1.25	
21					24.2	0.743	529	725	1.37	
22			104.9	343	21.1	0.345	859	1010	1.18	
23					24.2	0.349	892	1090	1.22	
24			231.1		21.1	0.760	682	801	1.17	
25					24.2	0.769	699	867	1.24	
26	25.4		106.7	524	27.9	2.005	11	15	1.36	
27	38.1					1.379	73	110	1.50	
28	50.8					0.990	114	121	1.06	
29	76.2				27.3	0.642	265	320	1.21	
30	127.3		50.0	371	66.2	0.198	1115	1285	1.15	
31				329		0.196	1089	1285	1.18	
32	127.0		50.8	371		0.195	1267	1303	1.03	
33				329		0.191	1218	1303	1.07	
34	101.6		152.4	605	34.1	0.752	641	818	1.28	
35							641	801	1.25	
36	120.9		104.9	452	24.4	0.395	981	1157	1.18	
37					29.6	0.389	934	1094	1.17	
38					25.9	0.385	896	952	1.06	
39	152.4		227.1	415	20.9	0.622	827	939	1.13	
40							827	881	1.06	
41	76.5		152.4	364	25.0	0.759	197	245	1.24	
42			61.0		40.9	0.330	321	423	1.32	
43					25.9	0.305	257	330	1.28	

TABLE 1 Continued

No.	ϕ (mm)	t (mm)	kL (cm)	σ_y (N/mm ²)	f_c (N/mm ²)	$\bar{\lambda}$	N_{cal} (kN)	N_{test} (kN)	$\frac{N_{test}}{N_{cal}}$	Ref.
44	114.3		83.8	414	28.9	0.319	719	734	1.02	
45							719	756	1.05	
46	127.0		149.9	289	35.2	0.489	644	636	0.99	
47							644	623	0.97	
48							644	658	1.02	
49	152.4		149.9	330	21.0	0.388	605	681	1.13	
50					25.9	0.404	685	725	1.06	
51							685	734	1.07	
52	168.1		81.3	298	43.3	0.205	1636	2233	1.36	
53							1636	2113	1.29	
54	88.9		172.7	400	40.0	0.870	554	614	1.11	
55			142.2		39.6	0.712	624	712	1.14	
56			111.8		39.1	0.562	692	716	1.03	
57			81.3		41.8	0.411	763	916	1.20	
58			50.8		40.8	0.256	804	992	1.23	
59	88.9		172.7	482	41.4	0.987	241	225	0.93	
60			142.2		37.0	0.802	271	294	1.08	
61			111.8		40.8	0.639	324	356	1.10	
62			81.3			0.465	355	400	1.13	
63			50.8			0.291	380	489	1.29	

ϕ : diameter of circular hollow section, t : wall thickness.

$\frac{N_{test}}{N_{cal}}$: $n = 100$, $\bar{X} = 1.135$, $S = 0.127$.

n : number of specimens, $\bar{X}$: mean value, S : standard deviation.

N_y is calculated by eqn (6'). In Fig. 4, they are compared with the formulated reduction coefficient, g , given by eqn (5). From this figure, it can be observed that eqn (6') can be applied without modification for the estimation of the squash load, N_y , i.e. the effect of confinement of concrete is insignificant, and that the curve b can be used for the estimation of the reduction coefficient, g . The test results, N_{test} , and the calculated column strength, N_{cal} , using $1.0 f_c$ and curve b are compared in Table 2.

3 CONCRETE-ENCASED STEEL COLUMN

The squash load, N_y , and the relative slenderness, $\bar{\lambda}$, are given by eqn (6) and eqn (7), respectively. Only four test results of specimens with longitudinal reinforcements from Japan⁸ and 30 test results of specimens without longitudinal reinforcements by Anslijn & Janss⁹ are available (see Fig. 5). All test

TABLE 2
Concrete-filled Steel Tubular Columns. Square Hollow Sections ($1.0 f_c$)

No.	B (mm)	t (mm)	kL (cm)	σ_y (N/mm^2)	f_c (N/mm^2)	$\bar{\lambda}$	N_{cal}	N_{test}	$\frac{N_{test}}{N_{cal}}$	Ref.
1	149.8	4.3	59.9	412	32.0	0.151	1639	1598	0.97	6
2			119.8		31.9	0.302	1578	1586	1.01	
3			179.8		32.8	0.455	1495	1573	1.05	
4			269.6		30.3	0.674	1281	1356	1.06	
5			359.5		32.2	0.906	1080	1143	1.06	
6	60.2	1.5	449.4	431	32.2	1.325	848	909	1.07	7
7			25.0		25.5	0.154	231	238	1.03	
8			50.0			0.309	222	227	1.02	
9			75.0			0.463	208	217	1.04	
10			100.0			0.620	191	192	1.00	
11			125.0			0.770	171	193	1.13	
12			150.0		24.0	0.920	146	157	1.07	
13			175.0		24.0	1.080	124	146	1.18	

B : width of square hollow section, t : wall thickness.

$\frac{N_{test}}{N_{cal}}$: $n = 13$, $\bar{X} = 1.053$, $S = 0.053$.

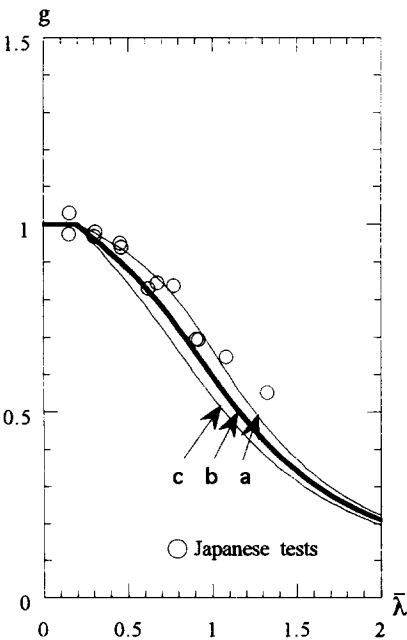


Fig. 4. Concrete-filled square steel tube columns, $1.0 f_c$.

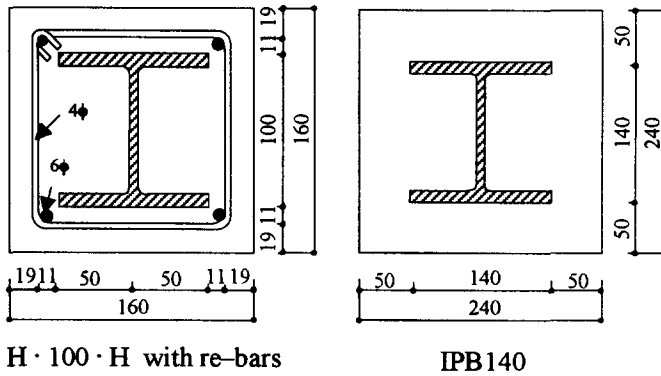


Fig. 5. Concrete-encased steel sections.

specimens buckled about the weak axis of the steel H-section. The variation of material properties covered are 269–380 N/mm² in yield stress of steel and 18.5–38.0 N/mm² in compressive strength of concrete in major parts of these test specimens. But some special test specimens using high yield stress (488–502 N/mm² in yield stress) and regular concrete (31.4–38.0 N/mm² in compressive strength) were also tested. For test specimens on which the modulus of elasticity of concrete, E_c , were not measured, E_c was estimated by eqn (8).

First of all, the experimental reduction coefficients, g_e , were estimated by

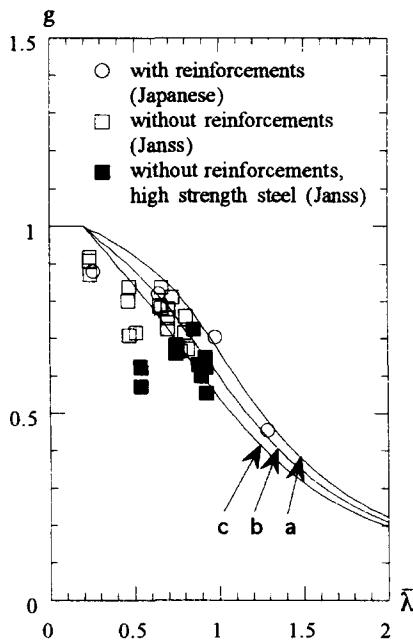


Fig. 6. Concrete-encased steel columns, 1.0 f_c .

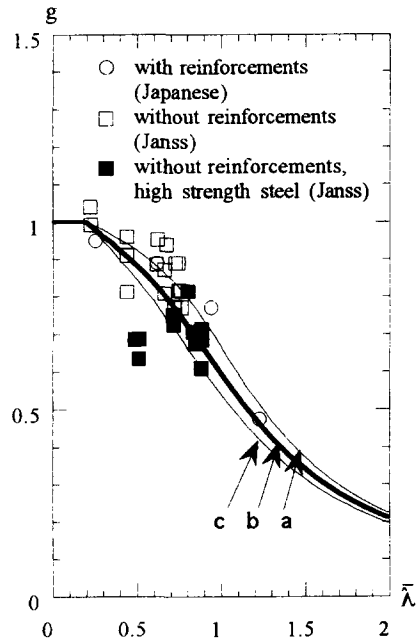


Fig. 7. Concrete-encased steel columns, $0.8 f_c$.

eqn (9), in which N_y is calculated by eqn (6). In Fig. 6, they are compared with the formulated reduction coefficient, g , given by eqn (5). From this figure it can be observed that actual squash load is lower than the squash load given by eqn (6). From the behavioural viewpoint, it can be suspected that the periphery zone of the concrete section had cracked prior to the occurrence of buckling. Through a parametric study, similar to the previous case, better fitting can be obtained using the reduced concrete strength, $0.8 f_c$, for the estimation of squash load, and adopting curve b for the reduction coefficient. The comparison of the reduction mode thus modified is given in Fig. 7, and

TABLE 3
Concrete-encased Steel Columns ($0.8 f_c$). With Longitudinal Reinforcements (4–6 mm ϕ , $\sigma_{yr} = 376 \text{ N/mm}^2$)

No.	Steel shape	Concrete (cm $\times$ cm)	kL (cm)	σ_y (N/mm ²)	f_c (N/mm ²)	$\bar{\lambda}$	N_{cal} (kN)	N_{test} (kN)	$\frac{N_{test}}{N_{cal}}$	Ref.
1	H.100	16 $\times$ 16	92.4	306	18.5	0.250	1030	996	0.97	8
2	100.6.8		230.9	298	21.4	0.618	902	974	1.08	
3			346.4	304	22.5	0.937	729	874	1.20	
4			461.9	304	20.6	1.228	487	522	1.07	

the correlation between test results, N_{test} , and the calculated column strength, N_{cal} , using $0.8 f_c$ and curve b is shown in Tables 3 and 4. Both from this figure and the tables, it should be noted that the combination of high strength steel section and regular strength concrete encasement without longitudinal

TABLE 4
Concrete-encased Steel Columns ($0.8 f_c$). Without Longitudinal Reinforcements

No.	Steel shape	Concrete ($cm \times cm$)	kL (cm)	σ_y (N/mm^2)	f_c (N/mm^2)	$\bar{\lambda}$	N_{cal} (kN)	N_{test} (kN)	$\frac{N_{test}}{N_{cal}}$	Ref.
1	IPE220	21×32	348.7	273	29.2	0.676	1951	2285	1.17	9
2			348.7	270	37.6	0.745	2169	2530	1.17	
3			348.7	269	33.7	0.720	2056	2059	1.00	
4			348.5	488	33.6	0.844	2376	2305	0.97	
5			348.5	488	31.9	0.802	2407	2707	1.12	
6			348.3	492	32.0	0.830	2361	2363	1.00	
7	IPB140	24×24	428.0	285	38.0	0.763	2087	2148	1.03	
8			428.1	285	35.1	0.759	2002	2177	1.09	
9			427.6	274	33.6	0.749	1969	2089	1.06	
10			348.6	293	33.6	0.727	2035	2344	1.15	
11			347.1	293	29.2	0.614	2039	2177	1.07	
12			347.8	293	435.1	0.621	2235	2579	1.15	
13			249.0	276	37.6	0.440	2495	2628	1.05	
14			247.6	276	38.0	0.439	2508	2236	0.90	
15			248.8	276	33.6	0.437	2342	2344	1.00	
16			128.8	276	33.6	0.226	2534	2550	1.01	
17			128.2	276	29.2	0.223	2364	2471	1.04	
18			125.3	276	35.4	0.221	2621	2746	1.05	
19			349.1	380	31.4	0.662	2338	2354	1.01	
20			348.3	380	33.7	0.662	2425	2628	1.08	
21			348.5	380	32.0	0.662	2359	2471	1.05	
22			427.4	502	31.4	0.881	2326	2354	1.01	
23			427.5	502	33.7	0.882	2377	2157	0.91	
24			427.4	502	32.1	0.881	2340	2481	1.06	
25			348.9	488	31.9	0.714	2647	2471	0.93	
26			348.8	488	33.7	0.714	2705	2618	0.97	
27			348.7	488	32.1	0.713	2663	2569	0.96	
28			248.5	500	33.6	0.485	3150	2432	0.77	
29			248.4	500	38.0	0.512	3277	2363	0.72	
30			248.8	500	37.6	0.512	3262	2550	0.78	

Regular steel, $\frac{N_{test}}{N_{cal}}$: $n = 22$, $\bar{X} = 1.064$, $S = 0.068$.

High strength steel, $\frac{N_{test}}{N_{cal}}$: $n = 12$, $\bar{X} = 0.933$, $S = 0.116$.

reinforcements will result in a risky prediction, especially in short columns. They are shown in bold face in Table 4.

4 SUGGESTED DESIGN FORMULAE

From the foregoing survey, the following design formulae can be suggested.

4.1 Concrete-filled steel tubular columns

$$N_c = gN_y(1)$$

$$N_y = A_s\sigma_y + \beta A_c f_c + (a_r\sigma_{yr}) \quad (10)$$

$$\beta = \begin{cases} 1.1 & \text{for concrete-filled circular tube columns} \\ 1.0 & \text{for concrete-filled square tube columns} \end{cases}$$

$$g = B(1 - \sqrt{1 - C}) \quad (11)$$

$$B = \frac{1 + 0.34(\bar{\lambda} - 0.2) + \bar{\lambda}^2}{2\bar{\lambda}^2}, \quad C = (B\bar{\lambda})^{-2}$$

$$\bar{\lambda} = \frac{kL}{\pi} \sqrt{\frac{A_s\sigma_y + \beta A_c f_c}{EI_s + E_c I_c}}. \quad (12)$$

4.2 Concrete-encased steel columns

Replacing N_y with N_y from eqn (13) and replace $\bar{\lambda}$ with $\bar{\lambda}$ from eqn (14) gives:

$$N_y = A_s\sigma_y + (a_r\sigma_{yr}) + 0.8A_c f_c \quad (13)$$

$$\bar{\lambda} = \frac{kL}{\pi} \sqrt{\frac{A_s\sigma_y + (a_r\sigma_{yr}) + 0.8A_c f_c}{EI_s + (EI_r) + E_c I_c}}. \quad (14)$$

The combination of high strength steel section and regular strength concrete for encasement should be avoided.

5 CONCLUSIONS

The basic approach for assessing the buckling strength of steel-concrete composite columns is quite the same as that of Eurocode 4¹⁰ or ECCS Recommendations.¹¹ But in these rules, the effective concrete strength for each type of composite column for assessing the squash load and the selection of buckling curves (*a*, *b*, *c*) for composite columns are not clearly specified. The following findings obtained from this study will improve the design practice of composite columns:

- (1) For the reduction coefficient, curve *b* of ISO standard or Eurocode 4 could be used for all types of composite columns.
- (2) In assessing the squash load, N_y , by eqn (10), β could be taken as $\beta = 1.1$ for concrete-filled circular tube columns; $\beta = 1.0$ for concrete-filled square tube columns; $\beta = 0.8$ for concrete-encased steel columns.

For concrete-encased steel columns, the combination of high strength steel section and regular strength concrete for encasement can not be recommended.

REFERENCES

1. ISO/DIS 10721, Steel structures—materials and design (this draft was approved on 2 December 1994 as an international standard).
2. CFT Working Group, Compression test on concrete-filled circular steel tube columns. Kozai Club, December 1993 (in Japanese).
3. Masuo, K. *et al.*, Buckling behavior of concrete-filled circular steel tubular columns using lightweight concrete. Proc. of the International Conference on Composite Construction, ASCCS, Fukuoka, Japan, 1991, pp. 95–100.
4. Sasaki, R., Experimental study on buckling strength of composite columns having steel tube. Annual Assembly of Architectural Institute of Japan, October 1984, pp. 2735–2736 (in Japanese).
5. Task Group 20, SSRC, a specification for the design of steel-concrete composite columns. *Engineering Journal*, AISC, 4th Quarter/1979, pp. 101–145.
6. Matsui, C., Stability and post-buckling behavior of concrete-filled square tubular columns. *Journal of Structural Engineering*, **39** (1993) pp. 323–334.
7. Nakamura, T., Experimental study on compression strength of concrete-filled square tubular steel columns. *Journal of Structural Engineering*, **40B** (1994) 411–417.
8. Matsui, C. *et al.*, Study on elastic-plastic behavior of concrete-encased columns subjected to eccentric axial thrust. Annual Assembly of Architectural Institute of Japan, September 1979, pp. 1627–1628 (in Japanese).
9. Anslijn, R. and Janss, J., Le calcul de charges ultimes des colonnes metalliques enrobees de beton. C.R.I.F., Report MT89, April 1974, Brussels.
10. Eurocode No. 4, Common unified rules for composite steel and concrete structures, ENV 1994-1-1.
11. ECCS Recommendations for composite structures, Construction Press.